

5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

- Your task is to:
- Create a stacked area plot using the data.
 - Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
 - Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

1

import matplotlib.pyplot as plt

2

import pandas as pd

3

4

Data for Months and Temperature for three cities

5

data = {

6

'Month': ['January', 'February', 'March', 'April', 'May', 'June',

'July', 'August', 'September', 'October', 'November', 'December'],

7

'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],

8

'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],

9

'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]

10

}

11

12

Write your code...

Average time

0.171 s

171.00 ms

Maximum time

0.171 s

171.00 ms

1 out of 1 shown test case(s) passed

Test case 1

171 ms

Debug

Expected output

Actual output

Temperature Variation

50

40

30

ature

Temperature Variation

50

40

30

ature

Terminal

Test cases

5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

- The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:
- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
 - `Gender`: Gender of the passenger (male/female).
 - `Age`: Age of the passenger.
 - `Survived`: Survival status (0 = Did not survive, 1 = Survived).
 - `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Explorer

titanicDat...

Submit

1 # import pandas as pd

2 # import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset from the CSV file

5 # df = pd.read_csv('titanic.csv')

6

7 # Set up the figure for 5 subplots

8 # fig, axes = plt.subplots(3, 2, figsize=(12, 12))

9

10 # write the code..

11 import pandas as pd

12 import matplotlib.pyplot as plt

13

Average time

0.645 s

645.00 ms

Maximum time

0.645 s

645.00 ms

1 out of 1 shown test case(s) passed

Test case 1

645 ms

Debug

Expected output

Actual output

Passenger Class Distribution

Gender Distribution

Age Distribution

Survival Count

Passenger Class Distribution

Gender Distribution

Age Distribution

Survival Count

Terminal

Test cases

5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- 1. Use **30 bins** for the histogram.
- 2. Set the **edge color** of the bars to **black (k)**.
- 3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
- 4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Histogram...

Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

Average time
0.210 s
210.00 ms

Maximum time
0.210 s
210.00 ms

1 out of 1 shown test case(s) passed

Test case 1 210 ms Debug

Expected output

Actual output

Age Distribution

Age Distribution

Terminal

Test cases

5.2.3. Bar plot of survival rate of passengers

08.36

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the **'Survived'** column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to **'bar'**.
3. Add the title **"Survival Count"** to the chart.
4. Label the x-axis as **'Survived'** and the y-axis as **'Count'**.

The Titanic dataset contains columns as shown below,

Pas sen ger id	Sur vive d	Pcla ss	Na me	Sex	Age	Sib Sp	Par ch	Tick et	Far e	Cab in	Em bark ed

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BarPlotOf...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time 0.263 s 263.00 ms

Maximum time 0.263 s 263.00 ms

1 out of 1 shown test case(s) passed

Test case 1 263 ms Debug

Expected output

Actual output

Survival Count

Survival Count

Terminal Test cases

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- 1. Group the data by the 'Sex' column, then use the value_counts() function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- 2. Use a stacked bar chart to display the survival counts.
- 3. Add the title "Survival by Gender" to the chart.
- 4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
- 5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

Average time0.276 s276.00 ms

Maximum time0.276 s276.00 ms

1 out of 1 shown test case(s) passed

Test case 1276 ms

Debug

Expected output

Actual output

Survival by Gender

Survival by Gender

Terminal

Test cases

5.2.5. Bar Plot for Survival by Pclass

04:57

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **"Survival by Pclass"** to the chart.
- Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
- The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
```

BarPlotOf...

Submit

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time0.277 s277.00 ms

Maximum time0.277 s277.00 ms

1 out of 1 shown test case(s) passed

Test case 1271 ms

Debug

Expected output

Actual output

Survival by Pclass

Survival by Pclass

Terminal

Test cases

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

- 1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
- 2. Set the chart type to 'bar' and make it stacked.
- 3. Add the title "**Survival by Embarked** " to the chart.
- 4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
- 5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925..S

BarPlotOf...

1 # import pandas as pd

2 # import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 # data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 # data['Age'].fillna(data['Age'].median(), inplace=True)

9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 # data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time

0.302 s

302.00 ms

Maximum time

0.302 s

302.00 ms

1 out of 1 shown test case(s) passed

Test case 1 302 ms

Debug

Expected output

Actual output

Survival by Embarked

Survival by Embarked

Not Survived

Survived

Terminal

Test cases

5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- 1. Use the **Pclass** column to group the data for the boxplot.
- 2. Set the title of the plot to **"Age by Pclass"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

Average time

0.285 s

285.00 ms

Maximum time

0.285 s

285.00 ms

1 out of 1 shown test case(s) passed

Test case 1 285 ms

Debug

Expected output

Actual output

Age by Pclass

Age by Pclass

Terminal

Test cases

Prev

Reset

Submit

Next

5.2.8. Box Plot for Age by Survived

02:42

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

- 1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
- 2. Set the title of the plot to **"Age by Survival"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
```

BoxPlotF...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time
0.268 s
268.00 ms

Maximum time
0.268 s
268.00 ms

1 out of 1 shown test case(s) passed

Test case 1 268 ms Debug

Expected output

Actual output

Terminal Test cases

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

- 1. Use the **Pclass** column to group the data for the boxplot.
- 2. Set the title of the plot to **"Fare by Pclass"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

Average time
0.274 s
274.00 ms

Maximum time
0.274 s
274.00 ms

1 out of 1 shown test case(s) passed

Test case 1 274 ms Debug

Expected output

Actual output

Terminal

Test cases

5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

AgeFareS...

Submit

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time0.192 s192.00 ms

Maximum time0.192 s192.00 ms

1 out of 1 shown test case(s) passed

Test case 1192 msDebug

Expected output

Actual output

Age vs. Fare

Age vs. Fare

TerminalTest cases

5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

- 1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
- 2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
- 3. Set the title of the plot to **"Age vs. Fare by Survival"**.
- 4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,S
```

AgeFareS...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time

0.219 s

219.00 ms

Maximum time

0.219 s

219.00 ms

1 out of 1 shown test case(s) passed

Test case 1219 ms

Debug

Expected output

Actual output

Age vs. Fare by Survival

Age vs. Fare by Survival

Terminal

Test cases